AI Resume → Job Match Analyzer

Mini Project Documentation

Development Platform: Base44 AI

Domain: Artificial Intelligence & Machine Learning

Application Type: AI-Powered Web Application

Live Demo: https://attractive-resume-match-pulse.base44.app

1. Project Overview

AI Resume → Job Match Analyzer is an AI-powered web application designed to help students, freshers, and job seekers understand how well their resume matches a particular job role. The application analyzes a user's resume and compares it with a given job description. It identifies matching skills, missing skills, skill gaps, and areas where the resume can be improved. The project was developed as a mini project using Base44 AI, which was used to design and build the web application.

2. Problem Statement

Many students and job seekers apply for jobs without knowing whether their resume actually matches the requirements of the job. They may miss important skills mentioned in the job description, include unnecessary information, fail to identify skill gaps, or not know which skills they should learn next. There is a need for a simple and intelligent system that can compare a resume with a job description and provide meaningful recommendations.

3. Proposed Solution

The AI Resume → Job Match Analyzer provides an easy-to-use platform where users can upload or enter their resume and provide a job description. The application analyzes both and provides an overall Job Match Score, matching skills, missing skills, partial skill matches, skill gap analysis, resume improvement suggestions, personalized learning recommendations, and job role compatibility.

4. Key Features

1. Resume Analysis — analyzes technical skills, programming languages, education, projects, certifications, experience, tools, and technologies.
2. Job Description Analysis — identifies required skills, preferred skills, qualifications, experience, technologies, responsibilities, and keywords.
3. Job Match Score — provides an overall compatibility score between the resume and target job.
4. Matching Skills — identifies skills present in both the resume and job description.
5. Missing Skills — identifies important job skills not clearly present in the resume.
6. Partial Matches — identifies related skills where the resume demonstrates some but not all required experience.
7. Skill Gap Analysis — shows current skills, missing skills, and priority levels.
8. Resume Improvement Suggestions — recommends ways to improve project descriptions, keywords, skills, achievements, and summary.
9. Personalized Learning Roadmap — suggests a learning plan based on missing skills.
10. AI Career Assistant — provides career guidance based on the user's resume and job analysis.

5. How the Application Works

The overall workflow is: User → Upload/Enter Resume → Enter Job Description → AI Analysis → Resume Analysis + Job Analysis → Skill Comparison → Job Match Score → Matching/Missing/Partial Skills → Skill Gap Analysis → Resume Improvement Tips → Personalized Roadmap.

6. Application Modules

Home Page — introduces the application and its purpose.
Dashboard — provides an overview of analyses and job matches.
Resume Analysis — accepts the user's resume.
Job Analysis — accepts the target job description.
Match Analysis — displays compatibility score and comparison.
Skill Gap — displays missing and required skills.
Recommendations — provides resume and skill recommendations.
Learning Roadmap — creates a personalized learning plan.
Analysis History — allows users to review previous analyses.

7. Technology Used

Development Platform: Base44 AI.
AI/ML Concepts: Natural Language Processing, resume information extraction, keyword extraction, semantic matching, skill gap analysis, recommendation systems, and AI-assisted career guidance.
Frontend: Modern responsive web UI, dashboard components, interactive cards, charts, and progress indicators.
Backend/Data: Base44 platform capabilities for application logic, data handling, and AI-powered functionality.

8. Target Users

B.Tech students, college students, fresh graduates, job seekers, internship applicants, career switchers, and placement preparation students.

9. Advantages

Easy to use; saves time during resume analysis; identifies important skill gaps; provides job-specific recommendations; helps students prepare for placements; provides explainable analysis; creates personalized learning recommendations; and is accessible through a web browser.

10. Future Enhancements

Multiple job comparison; ATS score prediction; job recommendation; built-in resume builder; LinkedIn profile analysis; personalized interview preparation; progress tracking; and integration with specialized NLP models for improved semantic similarity and skill extraction.

11. Project Objectives

1. Develop an AI-powered resume analysis system.
2. Compare resumes with job descriptions.
3. Identify relevant and missing skills.
4. Calculate job compatibility.
5. Provide personalized resume improvement suggestions.
6. Help students identify career skill gaps.
7. Recommend a learning roadmap based on job requirements.
8. Demonstrate the practical use of AI in career development.

12. Example Analysis

Resume Skills: Python, Machine Learning, SQL, Pandas, NumPy, Scikit-learn, Flask, Git.

Job Requirements: Python, Machine Learning, SQL, Docker, AWS, MLOps, Scikit-learn.

Example AI Analysis: Overall Match: 78%. Matching Skills: Python, Machine Learning, SQL, Scikit-learn. Missing Skills: Docker, AWS, MLOps. Recommendation: Focus on Docker, AWS, and MLOps to improve compatibility.

13. Testing

The application should be tested using technical resumes, fresher resumes, internship resumes, different job descriptions, different skill sets, and short/long resumes. Testing should verify that the application provides meaningful results for different inputs and handles invalid or incomplete input appropriately.

14. Suggested GitHub Repository Structure

AI-Resume-Job-Match-Analyzer/
■■■ README.md
■■■ screenshots/
■ ■■■ home-page.png
■ ■■■ dashboard.png
■ ■■■ resume-analysis.png
■ ■■■ match-score.png
■ ■■■ skill-gap.png
■■■ documentation/
■ ■■■ project-documentation.pdf
■■■ demo/
■■■ project-link.txt

15. Getting Started

Since this project was developed using Base44 AI, users can access the deployed application directly through the live demo link. Open the application in a web browser and start analyzing a resume against a job description.

16. Conclusion

The AI Resume → Job Match Analyzer demonstrates how AI can be applied to solve a practical career-related problem. Instead of simply checking whether a resume contains certain keywords, the application provides users with a clearer understanding of their compatibility with a job, their missing skills, and the areas they can improve. The project also demonstrates how Base44 AI can be used to rapidly design and develop an AI-powered web application.

17. Project Details

Project Type: Mini Project
Domain: Artificial Intelligence & Machine Learning
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Live Demo: https://attractive-resume-match-pulse.base44.app

18. Acknowledgement

This project was developed as part of our learning journey in Artificial Intelligence and Machine Learning, with Base44 AI used as the application development platform.

19. Keywords

Artificial Intelligence, Machine Learning, NLP, Resume Analysis, Job Matching, Skill Gap Analysis, Career AI, AI Web Application, Base44 AI, B.Tech AI & ML, Job Recommendation, Resume Analyzer.